Please complete the test task to assess practical development skills in C++ and Qt.

Task: Audio Messenger

Topic: Development of a simple application for real-time voice transmission over the network.

Technology Stack (mandatory):

- Qt 5.15 (or higher within the Qt 5 series)
- C++17
- Build via qmake
- QUdpSocket
- QAudioInput + QAudioOutput

Functional Requirements

Develop a desktop application with a graphical user interface that allows audio to be transmitted between two instances of the program over UDP.

Main Features:

1. **“Settings” tab**
    - Fields for entering the local IP address and port (Local IP / Local Port)
    - Fields for entering the remote IP address and port (Remote IP / Remote Port)
    - Audio device selection:
        - Microphone (input device)
        - Speakers / headphones (output device)
    - “Apply settings” button
2. **“Call” tab**
    - “Start call” / “End call” buttons
    - Connection status indicator (green/red)
    - Display of incoming and outgoing audio volume levels (optional but desirable)
3. **Networking**
    - Data transmission and reception via QUdpSocket
    - Packet size ~320–640 bytes
    - Simple custom protocol over UDP (header + audio data)
4. **Audio**
    - Audio format: PCM 16-bit, 16000 Hz, Mono (8000 Hz is also acceptable)
    - Reading from the microphone → sending over UDP
    - Receiving over UDP → playback through speakers
    - Simple jitter buffer implementation (queue of 3–5 packets)

Mandatory Requirements

- The code must build via qmake (provide the complete .pro file)
- Use C++17 features
- Proper thread handling (QThread is recommended)
- Correct error handling
- Logging to console or a text field
- Clean, readable code with comments in **English**
- No memory leaks

What to Submit

1. Complete project source code (as a zip archive or a link to GitHub / GitLab)
2. AudioMessenger.pro file
3. README.md file containing:
    - Build and run instructions
    - Brief description of the architecture
    - Technical decisions made and their rationale
    - Any difficulties encountered (if any)

Evaluation Criteria

- Stable bidirectional audio transmission
- Quality of architecture and separation of responsibilities
- Correct use of Qt signals and slots
- Proper handling of network and audio errors
- Code style and readability
- Use of modern C++17 features
- Presence and correct operation of the jitter buffer

Deadline: 5–7 days from the moment the task is received.

If necessary, a shorter deadline can be agreed upon with a simplified version (without device selection, without tabs, etc.).

If you have any questions regarding the task — please write immediately.

Good luck!